

# Event-TimeRAF: AI Claim Verification Log

Verified against run 20260723T112033170131Z

**Evidence contract.** The executed verification notebook checks all 43 manifest artifacts by SHA-256, reruns saved predictors, reconstructs metrics from 172,776 test forecast points, reruns paired block-bootstrap ablations, and regenerates the manuscript figures. The immutable file `event_timeraf_final_run.zip` is the numerical source of truth.

| Claim                                                                                                                                                | Checked how                                                                                                                      | Result             |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Run artifacts are unchanged and attributable<br>48,332 windows; 34,020/7,113/7,199<br>chronological split<br>Context/model dimensions are 86 and 152 | Recomputed SHA-256 for all 43 files listed in the archived manifest                                                              | Confirmed          |
|                                                                                                                                                      | Reconstructed saved NPZ arrays and joined window metadata; asserted shapes and split counts                                      | Confirmed          |
|                                                                                                                                                      | Counted saved feature names and inspected the M09 model: 23+14+9+39=85; 85+51+6+9=151                                            | Corrected          |
| M04 reports MSE/MAE/RMSE/ $R^2$ of 26.185/3.125/5.117/0.379<br>Full M09 improves on M04                                                              | Executed archived M04 on the reconstructed test matrix and recomputed all four metrics                                           | Confirmed          |
|                                                                                                                                                      | Executed M09; MSE is 26.712 versus 26.185 for M04; paired M09–M04 interval is [−0.026, 1.182]                                    | Corrected          |
| Retrieval improves frozen Chronos-Bolt conclusively                                                                                                  | Rebuilt validation-selected fusion; MSE changes 28.941 to 28.709, but paired interval [−0.665, 0.152] crosses zero               | Corrected          |
| Component-ablation values are measured                                                                                                               | Reran eight prediction-level comparisons with 500-resample, 24-hour paired block bootstrap                                       | Confirmed          |
| M12 drift gating is a verified final result                                                                                                          | Checked manifest membership and hashes; M12 appears only in post-manifest edited artifacts and was removed from the paper        | Corrected          |
| Seeded random retrieval is bitwise portable                                                                                                          | Compared a NumPy 2.4.1 replay with NumPy 2.0.2 evidence; M05/A01 are reconstructed from checksum-verified evidence               | Flagged unverified |
| Storm-event context is available in strict real time                                                                                                 | Inspected NOAA audit fields; publication timestamps are absent, so event results are retrospective sensitivity analyses          | Flagged unverified |
| Findings generalize geographically and runtime is hardware-normalized                                                                                | Checked dataset inventory and manifest hardware fields; only Los Angeles County is evaluated and hardware identifiers are absent | Flagged unverified |

**AI writing removed or corrected (before → after).**

|                        |                                                         |                                                                                                            |
|------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Dimensions</b>      | “86-dimensional context; 40 event features; 152 inputs” | “85-dimensional context; 39 event features; 151 per-horizon inputs”                                        |
| <b>Headline result</b> | “M12 is the best final model”                           | M12 result removed; M04 is the strongest manifest-backed model                                             |
| <b>Drift method</b>    | Generic weighted drift-score equation                   | Five implemented raw statistics, validation median/MAD scaling, clipping, and the 0.90 validation quantile |
| <b>Claim scope</b>     | Event-aware findings phrased as operational evidence    | Retrospective event sensitivity, one-county validation, and non-normalized runtime stated explicitly       |